within the computer or some other kind of trick, limited the computer to simulating local physics, rendering it impossible to fully simulate quantum effects. But, Feynman speculated, there might be another way to get the job done. “Can you do it with a new kind of computer—a quantum computer?” Feynman wondered. “I’m not sure.… So I leave that open.”

Several years later, a young physicist named David Deutsch took up where Feynman had left off. In 1985, Deutsch proved that a quantum computer—a computer taking full advantage of the difference between quantum physics and classical physics—could perform tasks more efficiently than an ordinary classical computer. Deutsch’s proof opened up the possibility of practical technological applications for Bell’s ideas, a feat that Bell had never foreseen. But Deutsch didn’t actually provide an example of how a quantum computer could outperform a classical one—he had merely proven that it could be done in theory. Finding an algorithm for a computer that hadn’t been built, to outperform all existing ones, was a tall order.

Nearly a decade later, a brilliant mathematician named Peter Shor filled that order in spectacular fashion. In 1994, he devised a quantum algorithm that could rapidly factor extremely large numbers—an extraordinarily important result. Not only was this a true demonstration of what Deutsch had proven possible, but Shor’s algorithm had massive practical consequences. It’s difficult for ordinary computers to factor large numbers—and, as Shor well knew, this difficulty was the basis for nearly all forms of practical cryptography, especially for secure communications over the newly burgeoning Internet. Shor had demonstrated that any kind of secure financial transaction over a computer network—from buying books to trading stocks—would be impossible to accomplish by conventional means in a world with working quantum computers.

But, by that time, quantum information theory had also yielded a solution to this problem: quantum cryptography. In fact, two forms of absolutely secure communication had been devised based on work first done in quantum foundations. One method, developed by Charles Bennett and Giles Brassard in 1984, was based on a result known as the “no-cloning theorem,” which had been proven in response to work done by the Fundamental Fysiks Group. Another method, developed by Artur Ekert in 1991, was based directly on Bell’s theorem. Both held the promise of